

Lecture 20: Wrap-up

PPHA 34600
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Why are we here?

I had two main goals for this course:

- ➊ Introduce you to program evaluation, and build familiarity with modern methods
- ➋ Prepare you to be a *consumer* of program evaluation

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Today:

- A whirlwind tour back through the quarter

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Today:

- A whirlwind tour back through the quarter
- Economics tells us that people respond to incentives...

We spent most of the class thinking about research designs

We explored a variety of different approaches:

- Randomized controlled trials (RCTs)
- Trying to control for observable things
- Panel data
- Instrumental variables
- Regression discontinuity
- Big Data and machine learning

That pesky fundamental problem

What are we trying to deal with?:

- We have $i \in \{1, \dots, N\}$ units
- $D_i \in \{0, 1\}$ is the treatment indicator for unit i
 - Treated units: $D_i = 1$
 - Untreated units: $D_i = 0$
- $Y_i(D_i)$ is the outcome for unit i with treatment status D_i
- The **treatment effect** for unit i is just:

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... but we can never observe both $Y_i(1)$ and $Y_i(0)$ simultaneously!



Selection is an issue

There are lots of things we want to estimate

We need to get around selection bias to do this

In other words, we need:

$$E[Y_i(1)] = E[Y_i(1)|D_i = 1] = E[Y_i(1)|D_i = 0]$$

and

$$E[Y_i(0)] = E[Y_i(0)|D_i = 0] = E[Y_i(0)|D_i = 1]$$

Regression equivalent:

$$E[\varepsilon_i|D_i] = 0$$

Randomization solves our selection issue

The ATE is just:

$$\tau^{ATE} = E[Y_i(1)] - E[Y_i(0)]$$

Since we have random assignment, we can estimate this as:

$$\hat{\tau}^{ATE} = \overline{Y(1)} - \overline{Y(0)}$$

Regression is a convenient way to do this:

$$Y_i = \alpha + \tau D_i + \varepsilon_i$$

This is why the RCT is the gold standard against which we judge all other research designs

With non-compliance, we can't estimate everything

What can we estimate with non-compliance?

- **ITT:** $\bar{Y}(1) - \bar{Y}(0)$
- **LATE:** $\frac{\bar{Y}(1) - \bar{Y}(0)}{\pi^C}$
 - Under constant treatment effects: equal to ATE, ATT
 - With heterogeneous treatment effects: equal to ATT
 - With defiers: equal to 💀

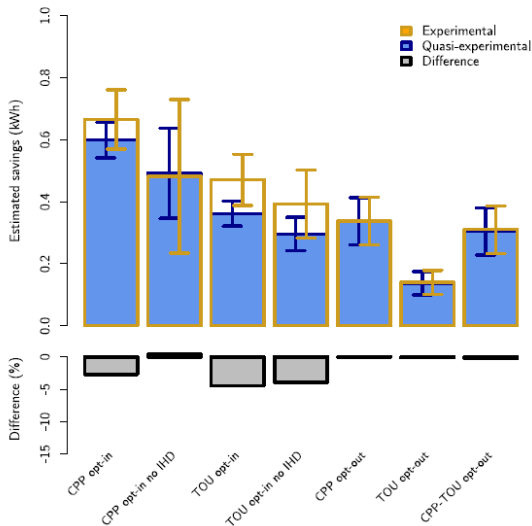
A randomized saturation design measures spillovers

- Start with N individuals who live in C (disjoint) clusters
- **Step 1:** Randomly assign clusters a treatment saturation $\pi_c \in [0, 1]$
 - Choose from a pre-determined set of saturations! (eg $\pi \in \{0.25, 0.75\}$)
 - It's common to have a pure control cluster: $\pi_c = 0$
- **Step 2:** In each cluster, randomly assign $\pi_c \cdot N_c$ units into treatment
 - Now D_{ic} is the treatment status for unit i in cluster c

This results in three types of units:

- Treated individual: $D_{ic} = 1$ as usual
 - Share in sample: μ
- Pure control: $D_{ic} = 0$ and $\pi_c = 0$
 - Share in sample: ψ
- Within-cluster control: $D_{ic} = 0$ and $\pi_c > 0$

We want to be cautious with non-experimental findings



Selection on observables is tricky

We took a first look at non-experimental estimation:

- SOO requires **extremely strong** assumptions (💀)
- ...but when you're stuck in this world, you can use:
 - 1 Regression adjustment
 - Controlling for stuff
 - 2 Matching
 - Pairing treated and untreated on observables

→ We typically **don't** believe SOO designs

Instrumental variables as selection on unobservables

We want to split $D_i = B_i\varepsilon_i + C_i$ into the C_i and other parts

An instrumental variable...:

...Generates variation in C_i but is uncorrelated with ε_i

Z_i is a valid instrument for D_i when the following are satisfied:

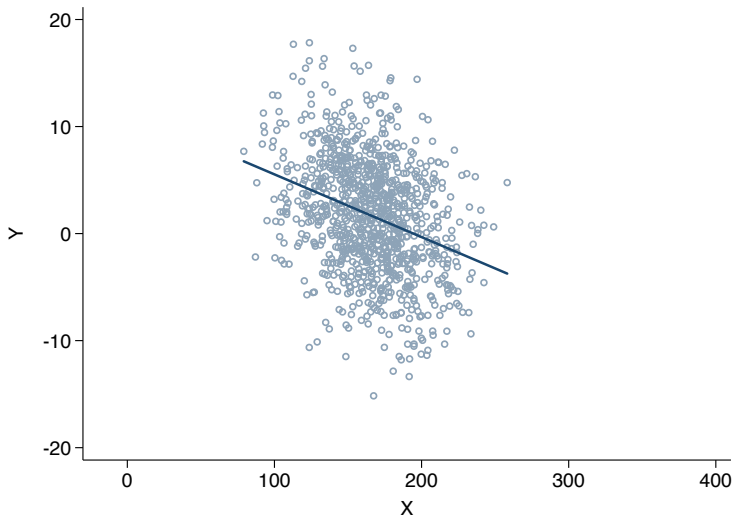
① **First stage:** $\text{Cov}(Z_i, D_i) \neq 0$

- Z_i and D_i are related
- Without this, you're capturing nothing
- This is actually testable!

② **Exclusion restriction:** $\text{Cov}(Z_i, \varepsilon_i) = 0$

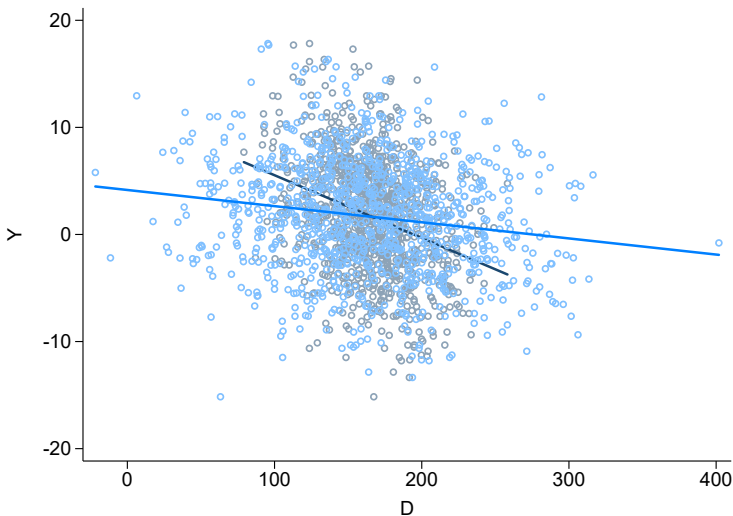
- Z_i and ε_i are **not** related
- Z_i only affects Y_i through D_i
- Fundamentally untestable! 💀

Classical measurement error in D_i is bad



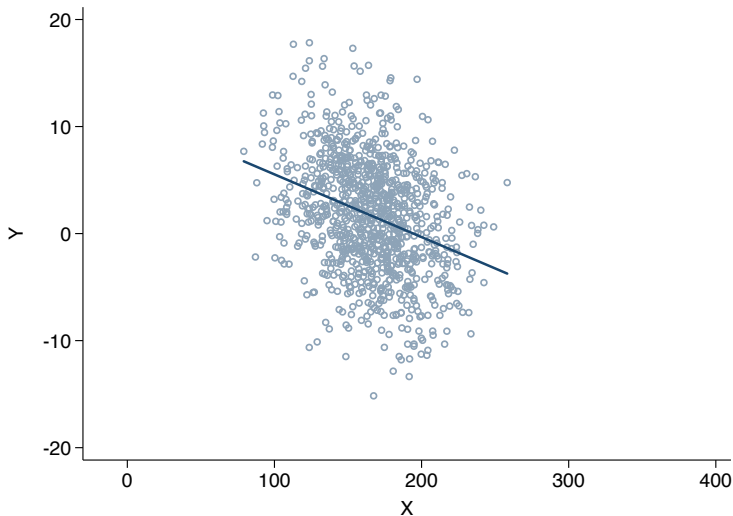
True relationship: $\tau = -0.059$

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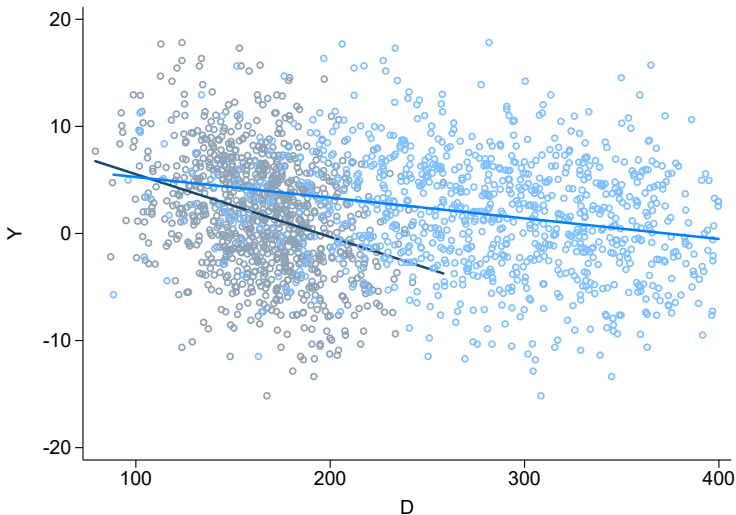
Estimated relationship: $\hat{\tau} = -0.015$

Non-classical measurement error in D_i is bad



True relationship: $\tau = -0.059$

Non-classical measurement error in D_i is bad



Estimated relationship: $\hat{\tau} = -0.019$

IV solves measurement error

What's the intuition?

- $\tilde{D}_i = D_i + \gamma_i$
 - $Z_i = \mathring{D}_i = D_i + \zeta_i$
- Z_i and $\tilde{D}_i$ only have the **true** D_i in common
- We've assumed that $\text{Cov}(\gamma_i, \zeta_i) = 0$

The first stage is:

$$\tilde{D}_i = \alpha + \pi \mathring{D}_i + \epsilon_i$$

- We're only using the variation from D_i (not from ζ_i or γ_i)!

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$$\tilde{D}_i = \alpha + \pi \check{D}_i + \epsilon_i$$

- We're only using the variation from D_i (not from ζ_i or γ_i)!
- **Important caveat:** This does not work with binary D_i !
- If true $D_i = 1$, measurement error can only be -1 or 0
 - If true $D_i = 0$, measurement error can only be 0 or 1
- Measurement error in $\tilde{D}_i$ and $\check{D}_i$ will be correlated

With heterogeneous treatment effects, IV gives us LATE

$$\hat{\tau}^{IV} = E[Y_i(1) - Y_i(0) | D_i(1) > D_i(0)]$$

- $\hat{\tau}^{IV}$ estimates the (L)ATE, conditional on $D_i(1) > D_i(0)$
- $D_i(1) > D_i(0)$ means Z_i moves D_i

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- $D_i(1) > D_i(0)$ means Z_i moves D_i
- We can divide the world into three groups:
 - 1 $D_i(1) > D_i(0)$: Compliers
 - 2 $D_i(1) = D_i(0) = 1$: Always-takers
 - 3 $D_i(1) = D_i(0) = 0$: Never-takers

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 - 2 $D_i(1) = D_i(0) = 1$: Always-takers
 - 3 $D_i(1) = D_i(0) = 0$: Never-takers
- Note that Z_i doesn't affect D_i for never-takers or always-takers
- The instrument is useless for them
- We can't learn about their treatment effects!
- (They essentially have no first stage)
- We can estimate LATEs for compliers only

Taking stock of IV

Under the right assumptions, we can use IV for...

- Eliminating bias due to measurement error
- Eliminating bias due to omitted variables
- Eliminating bias due to simultaneity
- Translating from ITT to LATE
- Estimating (L)ATEs

The trick is satisfying the exclusion restriction!

Moving to panel data land...

We've spilled a lot of ink on the selection problem:

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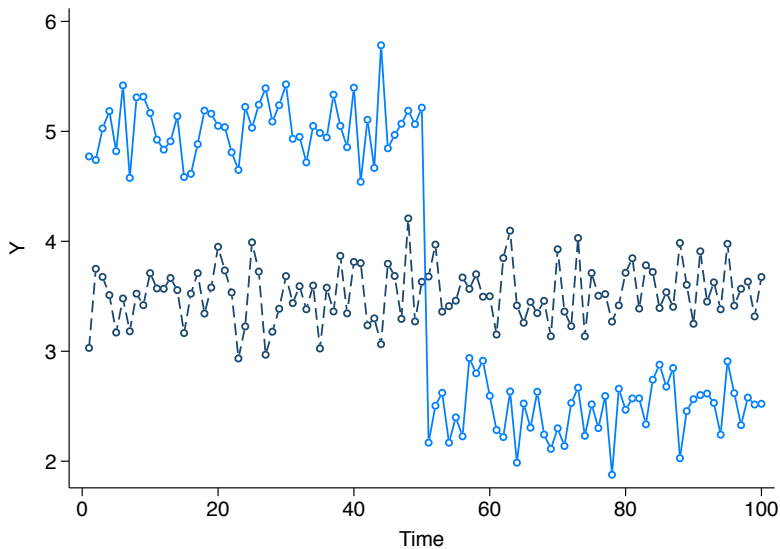
- To isolate the effect of D_i , we need potential outcomes to be the same among treated and untreated units
- With cross-sectional data, this is fundamentally tricky:
 - People, firms, households, etc are different from one another in lots of ways
 - Getting a clean comparison means separating τ from all of these differences

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- To isolate the effect of D_i , we need potential outcomes to be the same among treated and untreated units
- With cross-sectional data, this is fundamentally tricky:
 - People, firms, households, etc are different from one another in lots of ways
 - Getting a clean comparison means separating τ from all of these differences
- Enter time series data:
 - **Fundamental insight:**
Rather than comparing i to j , compare i in t to i in $t - 1$
 - In this formulation, i serves as a control for itself
 - i am much more similar to myself yesterday than i am to j

DD, visually



We can do this in regression form, too

The simplest implementation of DD is just:

$$\hat{\tau}^{DD} = (\bar{Y}(\text{treat}, \text{post}) - \bar{Y}(\text{treat}, \text{pre})) - (\bar{Y}(\text{untreat}, \text{post}) - \bar{Y}(\text{untreat}, \text{pre}))$$

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We can implement this via the following regression:

$$Y_i = \alpha + \tau \text{Treat} \times \text{Post}_{it} + \beta \text{Treat}_i + \delta \text{Post}_t + \varepsilon_{it}$$

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Running this regression yields $\hat{\tau} = \hat{\tau}^{DD}$

We can extend DD to the event study design

In the general version of this model, we include pre-treatment “effects”:

$$Y_{it} = \sum_{s=-R}^S \tau_s D_i \times \mathbf{1}[\text{periods to treatment} = s]_{it} + \alpha_i + \delta_t + \beta X_{it} + \varepsilon_{it}$$

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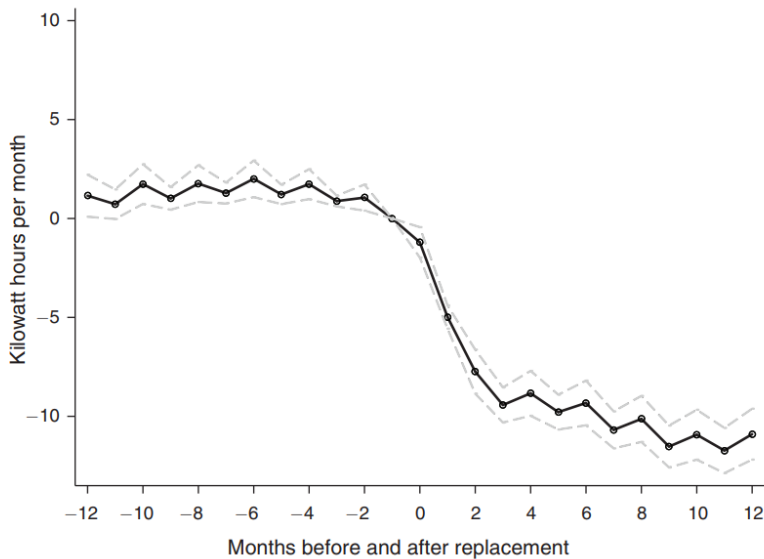
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- We get a (partial) test of the identifying assumption
 - We want pre-treatment τ_s 's to be centered on 0 and not trending

Event study, graphically



Sharp regression discontinuity

In the most straightforward, or “sharp” RD design:

- $Pr(D_i = 1|X_i \geq c) = 1$ and $Pr(D_i = 1|X_i < c) = 0$
- $Pr(D_i = 1|X_i \geq c) - Pr(D_i = 1|X_i < c) = 1$
- Nobody with $X_i < c$ gets treated
- Everybody with $X_i \geq c$ gets treated
- The probability of treatment jumps from 0 to 100% as X_i crosses c
- $D_i = \mathbf{1}(X_i \geq c)$

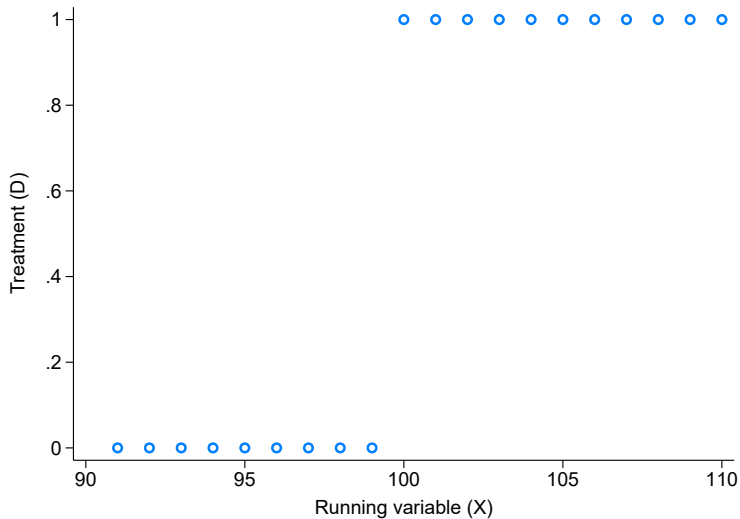
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→ This is equivalent to **perfect compliance** in the RCT

Sharp regression discontinuity: Treatment assignment



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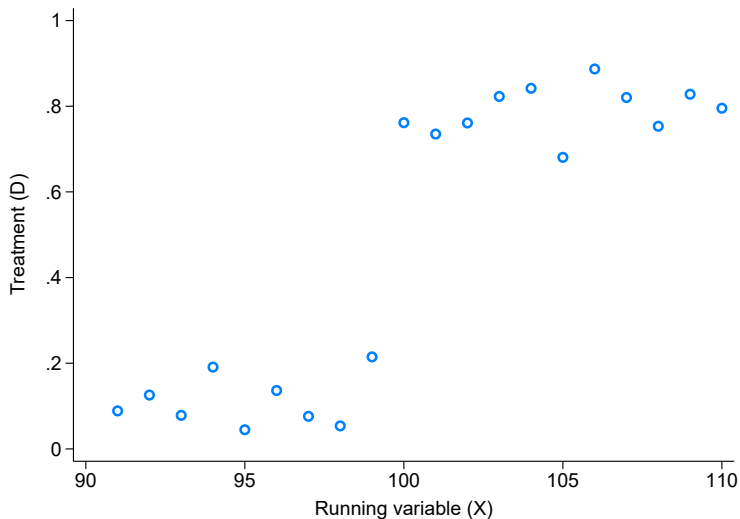
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Fuzzy regression discontinuity: Treatment assignment



Fuzzy RD: estimation methods

As with any other IV estimator, we can estimate fuzzy RD via 2SLS:

① **First stage:**

$$D_i = \alpha + \gamma \mathbf{1}[X_i \geq c] + \nu_i$$

② **Second stage:**

$$Y_i = \alpha + \tau \hat{D}_i + \varepsilon_i$$

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We can also estimate fuzzy RD using the first stage and reduced form:

➊ **First stage:**

$$D_i = \alpha + \gamma \mathbf{1}[X_i \geq c] + \nu_i$$

➋ **Reduced form:**

$$Y_i = \alpha + \theta \mathbf{1}[X_i \geq c] + \varepsilon_i$$

$$\hat{\tau}^{FRD} = \frac{\hat{\theta}}{\hat{\gamma}}$$

Big data can be a useful tool for program evaluation

- New data sources enable us to ask new questions
- We need to think carefully about data processing
 - Raw data
 - Processed in-house
 - Outsourced processing

Predicting $\hat{Y}$ without overfitting

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- Many ways to do this, but imagine running thousands of OLS regressions

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- Split the sample into pieces...
- Do step (1) on one part, and predict $\hat{Y}$ on the other part
- Record how well the model fits (eg $Y - \hat{Y}$)

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③ Repeat:

- Do this several times over different sample splits
- Pick the final model that does best

TL;DR:

- Randomized controlled trials (RCTs)
- Trying to control for observable things
- Panel data
- Instrumental variables
- Regression discontinuity
- Big Data and machine learning

What's next?

We've covered a lot in only 10 weeks! Possible next steps:

- Take more methods courses for an even deeper dive
- Take some topics classes to see these tools used
- Read the popular press to see these tools used (badly)
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Thank you for a fantastic quarter!